

Assignment 9

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Problem Statement

To use an appropriate algorithm and perform sentiment analysis for an Indian language.

Objective

The objective of this assignment is to analyze the sentiment of text written in Indian languages such as Hindi and Marathi using Natural Language Processing techniques and pre-trained models.

Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone behind a piece of text. It is widely used to classify text as positive, negative, or neutral. In this assignment, sentiment analysis is performed on Hindi and Marathi sentences.

Terminologies Used

1. Sentiment Analysis

Sentiment Analysis is the process of identifying and categorizing opinions expressed in text to determine whether the sentiment is positive, negative, or neutral.

2. Pre-trained Model

A pre-trained model is a machine learning model trained on large datasets and reused for specific tasks without training from scratch.

3. Multilingual Model

A multilingual model is capable of understanding and processing multiple languages, making it suitable for Indian language text.

4. Polarity

Polarity refers to the classification of sentiment into categories such as positive or negative.

Methodology

The following steps were performed:

1. A pre-trained sentiment analysis model was selected.
 2. Input text data was prepared in Hindi and Marathi.
 3. The model processed each sentence and predicted sentiment labels.
 4. The output included sentiment polarity and confidence score.
 5. The results were analyzed for correctness.
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Output Observation

It was observed that the default sentiment analysis model, which is primarily trained on English data, produced inaccurate results for Hindi and Marathi sentences. Many positive sentences were incorrectly classified as negative.

This highlights a limitation of using language-specific models on multilingual data.

Applications

- Social media analysis
- Customer feedback analysis
- Product review analysis

- Public opinion mining
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Conclusion

The assignment demonstrates sentiment analysis on Indian language text. However, results indicate that using an English-trained model leads to incorrect predictions. This emphasizes the need for multilingual or language-specific models for better accuracy.

Result

Sentiment analysis was successfully performed on Hindi and Marathi text, but the limitations of the model were observed, fulfilling the objectives of the assignment.